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Voltage-Mode Controlled Synchronous DC-DC Buck Converter using 0.13µ CMOS Switches

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Voltage-Mode Controlled Synchronous DC-DC Buck Converter using 0.13µ CMOS Switches

by

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Report

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Dedication

To my wonderful family:

Stephani and Elizabeth

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Abstract

Voltage-Mode Controlled Synchronous DC-DC Buck Converter using 0.13µ CMOS Switches

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This report is a study of the effects of a commercial 0.13µ process and automotive temperature corners on a synchronous DC-DC buck converter design. The basics of switching converters will be explored with an emphasis on voltage-mode controlled feedback. A Type-III compensation network is designed using transfer function analysis to compensate for the inherent double pole introduced by an LC network. The output of the compensation network will drive a pulse width modulation comparator to vary the duty cycle of the high-side PMOS and low-side NMOS transistor switches. After the synchronous buck converter design was complete, the effect of process and temperature on efficiency, output voltage ripple, inductor peak to peak current, and output voltage load response was examined.

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Chapter 1: Introduction

There are many mechanisms available to supply power to different circuitry in automotive electronic modules. Through the use of DC-DC converters, the typical 12 V car battery can be converted to a variety of voltage levels. The two main Types of DC-DC converters are linear and switching regulators. Linear regulators are based on the use of semiconductor devices (e.g. BJT or FET) operating in the linear (or active) region. Switching regulators use the same semiconductor devices, but operate them in the saturation region, essentially using the transistor as a switch. Linear regulators are popular in the automotive environment because of their low cost and low noise profile (i.e. low electromagnetic emissions). Unfortunately, this comes at a cost of poor efficiency (efficiency = input/output voltage) due to the fact that the semiconductor device has to regulate the output solely by introducing a voltage drop between the input and output nodes. The voltage drop is created by using the resistivity of the junction (drain and source in the case of a FET) of the transistor in the linear region. The power losses through the transistor are in the form of heat and must be effectively managed in high power linear regulator designs. Switching regulators on the other hand are complex devices that are inherently noisy (i.e. high electromagnetic emissions) due to the current flow through the transistor being turned on and off at the rate of the switching frequency. The main advantage of the switching regulator is that it has a much higher efficiency (typically 70 – 90%) which is especially important in battery operated designs. The ideal DC-DC converter for the automotive environment would have a low emissions profile, low cost, high efficiency, and small physical design.

To create a DC-DC converter with high efficiency, the best option is to use a switching regulator. When the output of the switching supply is less than the input voltage, this is referred to as a buck switching converter. This report will concentrate on designing a synchronous 3.3 V to 1.8 V buck converter, specifically using a commercial 0.13µ CMOS process for the low and high side transistor switches. The effect that process corners and automotive temperature corners have on common electrical and timing parameters will be studied and compared.

Chapter 2: Buck Switching Converter Design

2.1 BUCK CONVERTER BASICS

There are two main types of buck switching converters: asynchronous and synchronous (see Figure 1). Both types operate with the same basic functionality with the asynchronous buck converter switch and diode being analogous to the synchronous buck converter high-side switch (HSS) and low-side switch (LSS) respectively. During state one (on state), the switch and HSS are closed while the diode and LSS are reverse biased and open respectively. This introduces a voltage potential across the inductor of $V_L = V_{IN} - V_{OUT}$, if the voltage drop across switch or HSS is assumed to be zero. This potential gives rise to a linear current flow through the inductor according to $dI_L/dt = V_L / L$. The current in turn is used to store charge in the capacitor ($I = C * dV_{OUT}/dt$) as well as for the load circuitry. During the second state (off state), the switch and HSS are opened while the diode and LSS are forward biased and closed respectively. The voltage across the inductor changes to $V_L = V_{Diode} - V_{OUT}$ (asynchronous) or $V_L = V_{LSS} - V_{OUT}$ (synchronous). This causes a linear decrease in the current through the inductor. The major difference between these two buck converters is the power losses during the off state. The diode has a higher voltage drop (~200 mV) than the switch (~ 50 mV), which directly affects efficiency.

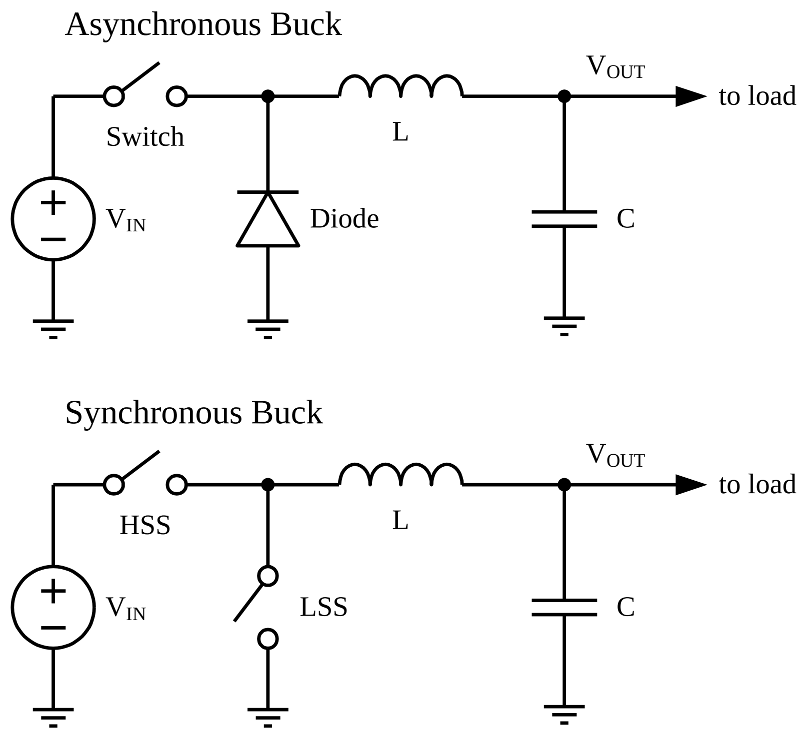

Figure 1: Buck Switching DC-DC Converter Types

To design a synchronous buck converter a control mechanism must be implemented to manage the HSS and LSS effectively. Also, since the output load will vary, this mechanism must adjust the duty cycle to meet sudden load changes while maintaining stability. There are two methods to accomplish this: voltage-mode control (VMC) and current-mode control (CMC). VMC senses the output voltage and response to any fluctuations by adjusting the duty cycle of the switches. The output voltage is typically sampled through a resistor divider that has a high enough impedance to draw a negligible amount of current. CMC senses current (through some means) at one of many options within the circuit while still using a resistor divider to sample the output voltage. When the sensed current is equal to the sampled voltage, then the duty cycle is regulated accordingly [1], [2]. The main disadvantage of VMC is that it must wait for a voltage change at the output before it can respond, whereas CMC will sense the current change before the output voltage change occurs. The drawback of CMC is that it adds an additional loop to the feedback network, making it even more complex to design. For this design, a VMC with stability compensation is implemented.

2.2 VOLTAGE-MODE CONTROL BUCK CONVERTER

A complete voltage-mode control buck converter diagram is shown in Figure 2. A compensation network is used to sample the V_{OUT} voltage and compare this to a known reference voltage (V_{REF}). Also during this stage, additional filtering is used to compensate the LC filter and produce a stable feedback network (see Section 2.6). The pulse width modulation (PWM) block consists of a comparator changes duty cycle at HSS and LSS to regulate a constant V_{OUT} over various load current conditions (see Section 2.3).

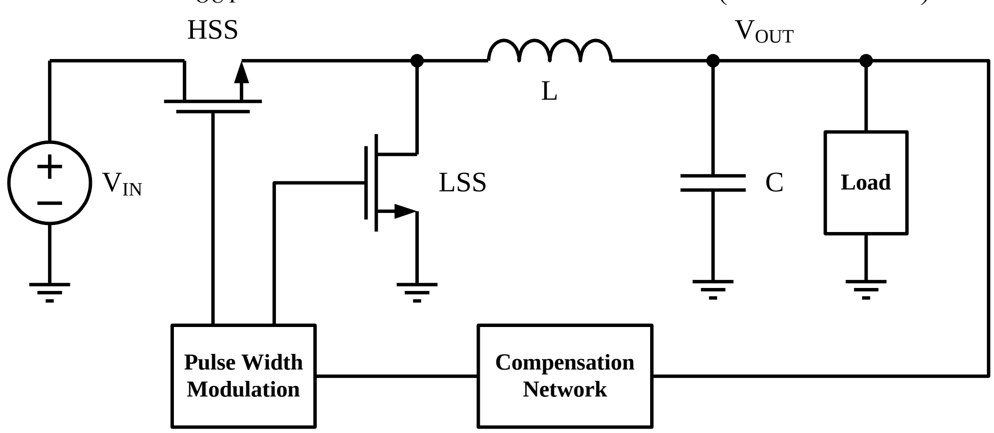

Figure 2: Voltage-Mode Control Synchronous Buck Converter

2.3 PULSE WIDTH MODULATION

The pulse width modulation circuit for this design involves using a comparator to compare a DC level (V_{ADJ}) to a triangle wave generator with amplitude V_{TW} (see Figure 3). The resulting output produces a square wave with a duty cycle based on the voltage level of V_{ADJ}. V_{ADJ} is derived from the Error Amplifier (see Figure 7) of the compensation network stage and is the difference between the reference voltage (V_{REF}) and the sampled V_{OUT} after a resistor divider (V_{ARD}). The resistor divider is sized so that the desired output voltage level is reduced to that of V_{REF}. The output states of the PWM are defined in Table 1. When the output voltage rises above the desired output voltage (e.g. the load current decreases), the PWM will introduce a lower duty cycle at the HSS and a higher duty cycle at the LSS to regulate to the desired output voltage. Alternatively, when the output voltage falls below the desired output voltage (e.g. the load current increases), the PWM will introduce a higher duty cycle at the HSS and a lower duty cycle at the LSS.

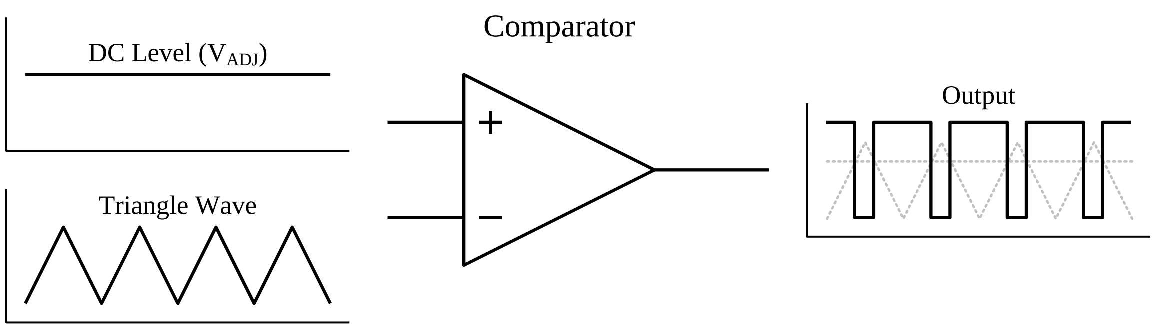

Figure 3: PWM Comparator

Error Amplifier Inputs	**V_{ADJ} Response**	**PWM Comparator Output Description**
$V_{ARD} > V_{REF}$	decreases	lower duty cycle at HSS, higher duty cycle at LSS
$V_{ARD} < V_{REF}$	increases	higher duty cycle at HSS, lower duty cycle at LSS
$V_{ARD} = V_{REF}$	no change	no change in duty cycle at HSS or LSS

Table 1: PWM Output States

In order to maintain the maximum efficiency through the PWM phase, the switching duty cycle of the HSS and LSS needs to be precisely aligned so that dead time (when neither HSS nor LSS is on) is reduced. Also, the state where both switches are on simultaneously must be avoided so that there is not a direct low impedance path to ground that will damage the transistors. For the LSS in this design, a second comparator is used to produce the same phase as that of the HSS input. To minimize the dead time and ensure that overlap does not occur, the triangle wave input of the HSS comparator is offset by a small voltage (V_{OFFSET}), see Figure 4. For clarity, the HSS comparator output is shown as the inverse of the actual polarity since it is driving a PMOS transistor. In real-world designs, a non-overlapping clock generator circuit drives the HSS and LSS transistors. This clock generator circuit would additionally manage the dead time and add any necessary adjustments to counteract process and temperature skews in the output phase.

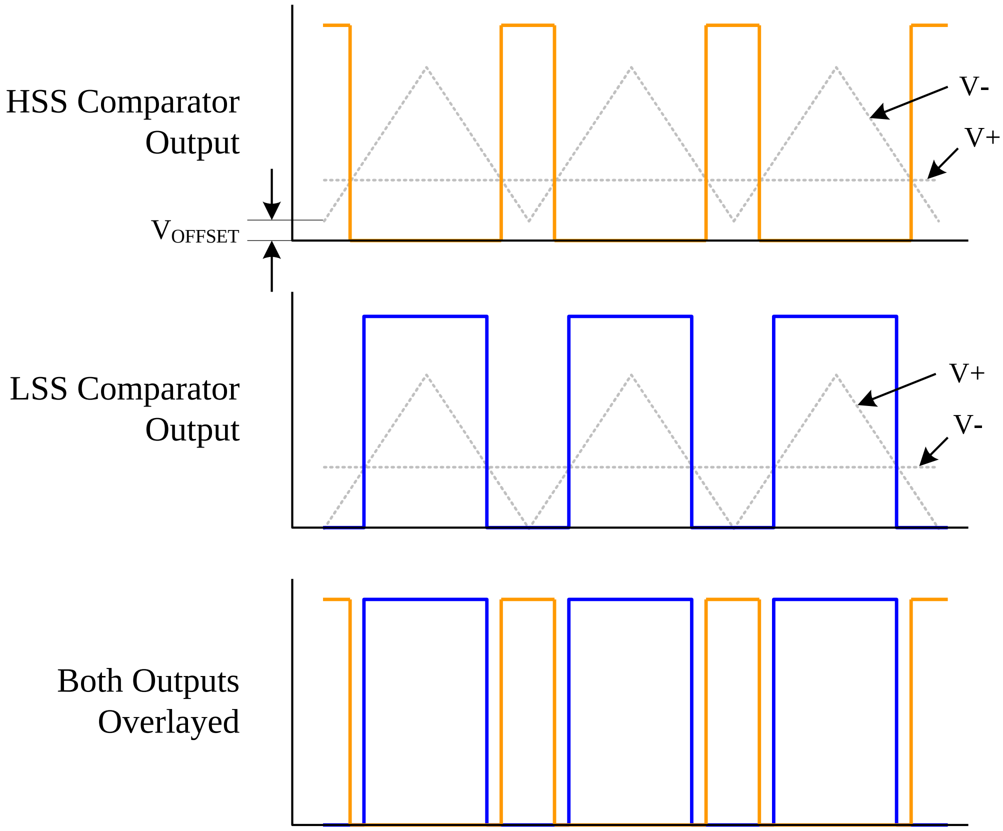

Figure 4: HSS and LSS Comparator Output Signals

2.4 Inductor Capacitor Filter Transfer Function

Before designing a compensation network, the current buck converter system needs to be analyzed. First, to effectively calculate the transfer function of the LC filter, the resistance of each component must be taken into account. Figure 5 includes the DC resistance of the inductor (R_L) along with the equivalent series resistance of the capacitor (R_C).

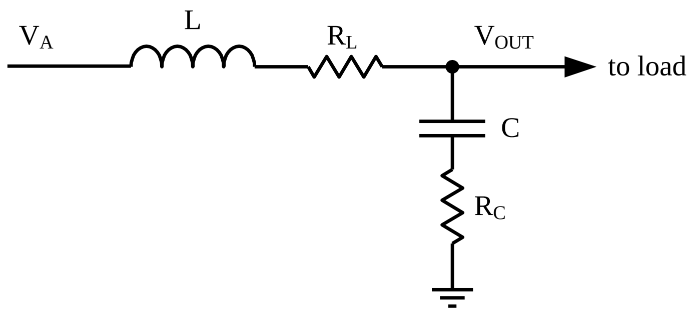

Figure 5: LC Filter with Component Resistance

The transfer function is calculated according to the equation below:

$$\mathrm{Gain}_{LC} = \frac{V_{OUT}}{V_A} = \frac{\frac{1}{sC} + R_C}{R_L + sL + \frac{1}{sC} + R_C} = \frac{1 + sR_C C}{1 + s(R_L + R_C)C + s^2 LC} \tag{1}$$

Gain_{LC} of the LC filter shows a second order filter with a single zero produced by the capacitance and its equivalent series resistance and a double pole produced by the resonant frequency of the inductance and capacitance. The frequency of the single zero and double pole are calculated below [3] and [4]:

$$F_{Z(R_C C)} = \frac{1}{2\pi R_C C} \tag{2}$$

$$F_{P(LC)} = \frac{1}{2\pi\sqrt{LC}} \tag{3}$$

The frequency at which the zero occurs ($F_{Z(R_C C)}$) is much greater than the frequency of the pole ($F_{P(LC)}$) due to the fact that R_C is in the mΩ range, while L is in the μH range.

2.5 OPEN LOOP GAIN

The open loop gain ($Gain_{OL}$) of the PWM and LC filter stages is calculated by multiplying the gain of the PWM stage ($Gain_{PWM}$) times the gain of the LC filter ($Gain_{LC}$). $Gain_{PWM}$ is simply V_{IN} divided by the voltage amplitude of the triangle waveform (V_{TW}).

$$Gain_{OL} = Gain_{PWM} \times Gain_{LC} = \frac{V_{IN}}{V_{TW}} \times \frac{1 + sR_C C}{1 + s(R_L + R_C)C + s^2 LC} \quad (4)$$

The bode plot of the open loop gain is shown in Figure 6. The unity gain frequency (F_0) is the frequency at which the magnitude of the gain is at 0 dB. For synchronous converters, F_0 should be typically placed between 15% and 30% of the switching frequency (F_{SW}). Comparing the phase angle at F_0 to 180 degrees will determine if the PWM and LC network is stable (called phase margin). The gain of the system can be increased but must ensure that the phase at F_0 does not exceed 180 degrees [4].

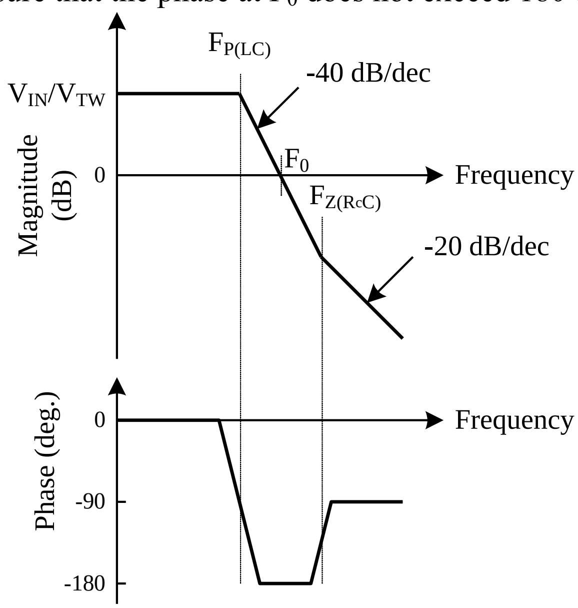

Figure 6: Bode Plot of Open Loop Gain

2.6 COMPENSATION NETWORK

Now that the open loop gain of the PWM and LC network is determined, a compensation filter can be designed so that the closed loop gain does not have a phase margin less than 45 degrees. Figure 6 shows that some compensation of the $F_{P(LC)}$ double pole is necessary to slow the -40 dB/decade roll-off. Ideally, the slope would be -20 dB/decade at F_0 to ensure a phase margin greater than 45 degrees. The recommended technique is to provide a phase boost to counteract the double pole. For this design, a Type-III compensation network will be used [5], [6], see Figure 7. The Type-III compensation network generates two zeros (introduces a 180 degree phase boost) to counteract the double pole.

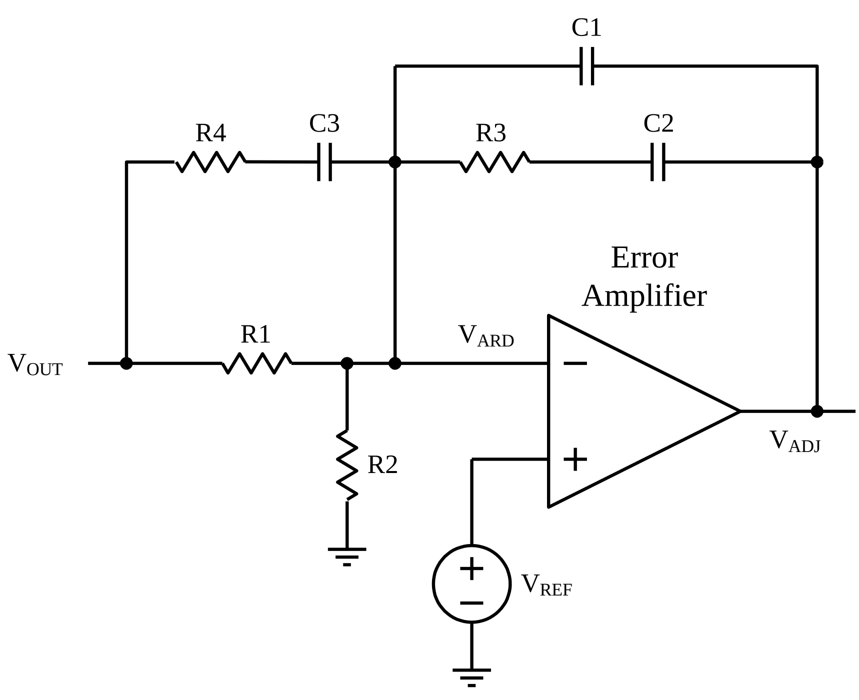

Figure 7: Error Amplifier with Type-III Compensation Network

The transfer function of the error amplifier with Type-III compensation network is given by:

$$H(s) = -\frac{V_{ADJ}}{V_{OUT}} = -\frac{(R3 + \frac{1}{sC2}) // \frac{1}{sC1}}{R1 // (R4 + \frac{1}{sC3})}$$
$$H(S) = -\frac{(1+sR3C2)(1+s(R1+R4)C3)}{sR1(C2+C1)(1+sR3(\frac{C2C1}{C2+C1}))(1+sR4C3)} \quad (5)$$

Assume the pole generated by C1 and R3 is set at a much higher frequency than the zero generated by C2 and R2 (therefore C2 >> C1). Thus the transfer function simplifies to:

$$H(s) = -\frac{(1+sR3C2)(1+s(R1+R4)C3)}{sR1C2(1+sR3C1)(1+sR4C3)} \quad (6)$$

The Type-III compensation network has two zeros and three poles at the following frequencies:

$$F_{Z(R3C2)} = \frac{1}{2\pi R3C2} \quad (7)$$

$$F_{Z(R1R4C3)} = \frac{1}{2\pi (R1+R4)C3} \quad (8)$$

$$F_{P(R1C2)} = 0 \quad (9)$$

$$F_{P(R4C3)} = \frac{1}{2\pi R4C3} \quad (10)$$

$$F_{P(R3C1)} = \frac{1}{2\pi R3C1} \quad (11)$$

2.7 COMPENSATION AND CLOSED LOOP BODE PLOT

The Bode plot for the compensation network and closed loop gain/phase is shown in Figure 8. The double zero introduced by the Type-III compensation network allows the resultant closed loop gain to have the desired -20 dB/decade at the unity gain frequency. Comparing the phase plot of the open loop system in Figure 6 to that of the closed loop system in Figure 8, it is evident that the compensation network has indeed created a more stable system by increasing the phase margin.

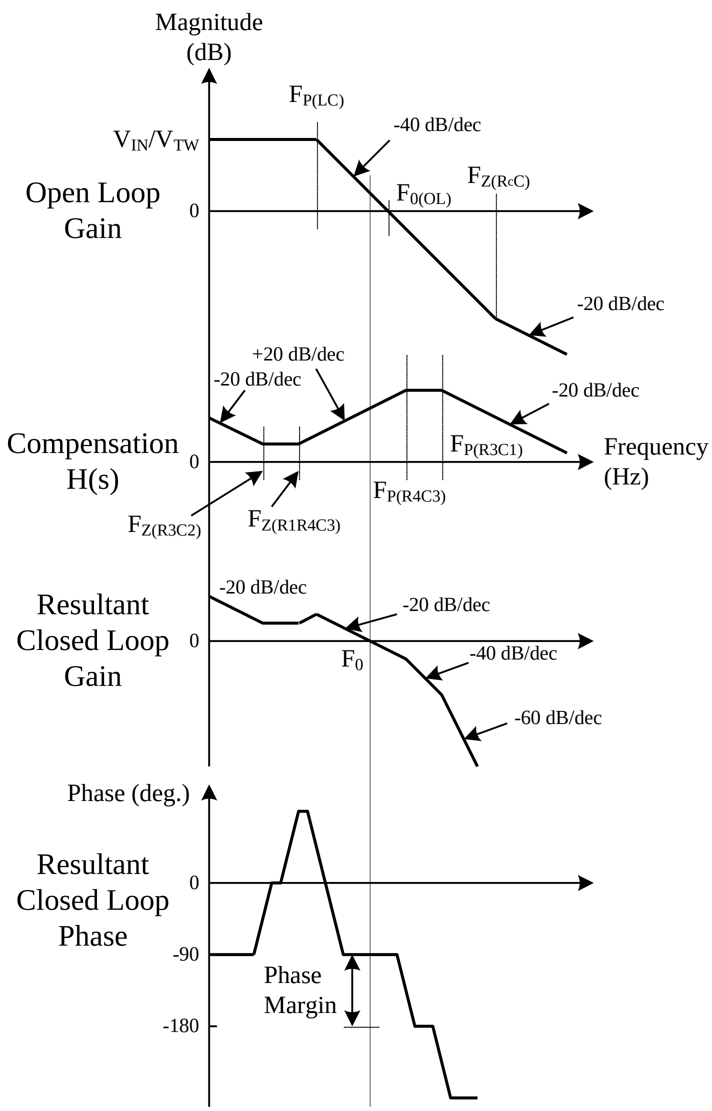

Figure 8: Bode Plot of Compensation Network and Closed Loop Gain/Phase

2.8 COMPENSATION POLE AND ZERO PLACEMENT

For a Type-III compensation network the R3 and C1 generated pole should be placed at half the switching frequency [6]:

$$F_{P(R3C1)} = \frac{F_{SW}}{2} \tag{12}$$

The zero generated by R1, R4, and C4 as well as the pole generated by R4 and C3 should be located at frequencies to produce the largest phase margin at F_0. These frequencies are calculated as follows:

$$F_{Z(R1R4C3)} = F_0\sqrt{\frac{1-\sin(70^\circ)}{1+\sin(70^\circ)}} \tag{13}$$

$$F_{P(R4C3)} = F_0\sqrt{\frac{1+\sin(70^\circ)}{1-\sin(70^\circ)}} \tag{14}$$

The zero generated by R3 and C2 is located at half the frequency of the zero generated by R1, R4, and C3:

$$F_{Z(R3C2)} = \frac{F_{Z(R1R4C3)}}{2} \tag{15}$$

By choosing a C3 value, the rest of the Type-III component values can be solved accordingly.

Using equation (10):

$$R4 = \frac{1}{2\pi C3 F_{P(R4C3)}} \tag{16}$$

Using equation (8):

$$R1 = \frac{1}{2\pi C3 F_{Z(R1R4C3)}} - R4 \tag{17}$$

R3 is calculated by manipulating the open loop gain (4), compensation transfer function (6), and the fact that the amplitude of the gain at the crossover frequency between (4) and (6) is equal to one.

$$R3 = \frac{2\pi F_0 LCV_{SW}}{V_{IN}C3} \tag{18}$$

Using equation (11):

$$C1 = \frac{1}{2\pi R3 F_{P(R3C1)}} \tag{19}$$

Using equation (7):

$$C2 = \frac{1}{2\pi R3 F_{Z(R3C2)}} \tag{20}$$

R2 is calculated by solving the resistor voltage divider formed between R1 and R2, V_{OUT} and V_{REF}:

$$R2 = \frac{R1 V_{REF}}{V_{OUT} - V_{REF}} \tag{21}$$

Chapter 3: Synchronous Buck Converter Results

3.1 CIRCUIT PARAMETERS

In order to calculate the component values of the compensation network, some parameters of the buck DC-DC converter must be defined (see Table 2).

Parameter	**Symbol**	**Value**
Temperature	Temp	-40 to 125 °C
Switching Frequency	F_{SW}	350 kHz
Input Voltage	V_{IN}	3.3 V
Output Voltage	V_{OUT}	1.8 V
Triangle Waveform Amplitude	V_{TW}	1.5 V
Triangle Waveform Offset Voltage	V_{OFFSET}	5 mV
Reference Voltage	V_{REF}	1.65 V
Output Load Current	I_{Load}	100 mA to 500 mA
Switching Inductor	L	10 μH
Inductor DC Resistance	R_L	5 mΩ
Inductor Current Limit	I_{LMAX}	1.0 A
Output Capacitor	C	4.7 μF
Capacitor Equivalent Series Resistance	R_C	2 mΩ

Table 2: Predefined Synchronous Buck Converter Parameters

The parameters predefined in Table 2 are then used to calculate the values of the Type-III compensation network (see Table 3). The resistor and capacitor values calculated are rounded to the nearest readily available standard values.

Parameter	Value	Equation
F_0	58.33 kHz	$16.66\% \times F_{SW}$
$F_{Z(R_C C)}$	16.93 MHz	(2)
$F_{Z(R1R4C3)}$	10.29 kHz	(13)
$F_{Z(R3C2)}$	5.145 kHz	(15)
$F_{P(LC)}$	23 kHz	(3)
$F_{P(R3C1)}$	175 kHz	(12)
$F_{P(R4C3)}$	330.81 kHz	(14)
C1	120 pF	(19)
C2	3.9 nF	(20)
C3	1 nF	common value chosen
R1	15 kΩ	(17)
R2	165 kΩ	(21)
R3	7.87 kΩ	(18)
R4	487 Ω	(16)

Table 3: Type-III Compensation Network Parameters

3.2 COMPLETE SCHEMATIC

The complete synchronous buck converter schematic using the predetermined and calculated component values is shown in Figure 9. The size of the LSS PMOS and HSS NMOS transistors was chosen to handle the maximum load current plus additional inductor peak current overhead. To size the HSS and LSS with respect to each other, an equivalent resistance was found using their respective duty cycles (D). For V_{IN} = 3.3 V and V_{OUT} = 1.8 V, the ideal duty cycle for the PMOS HSS (D_{HSS}) = 1.8 / 3.3 = 0.545, while the NMOS LSS (D_{LSS}) = 0.455. To size them correctly, $R_{LSS} = R_{HSS} * (D_{HSS} / D_{LSS})$ = 1.2 * R_{HSS}.

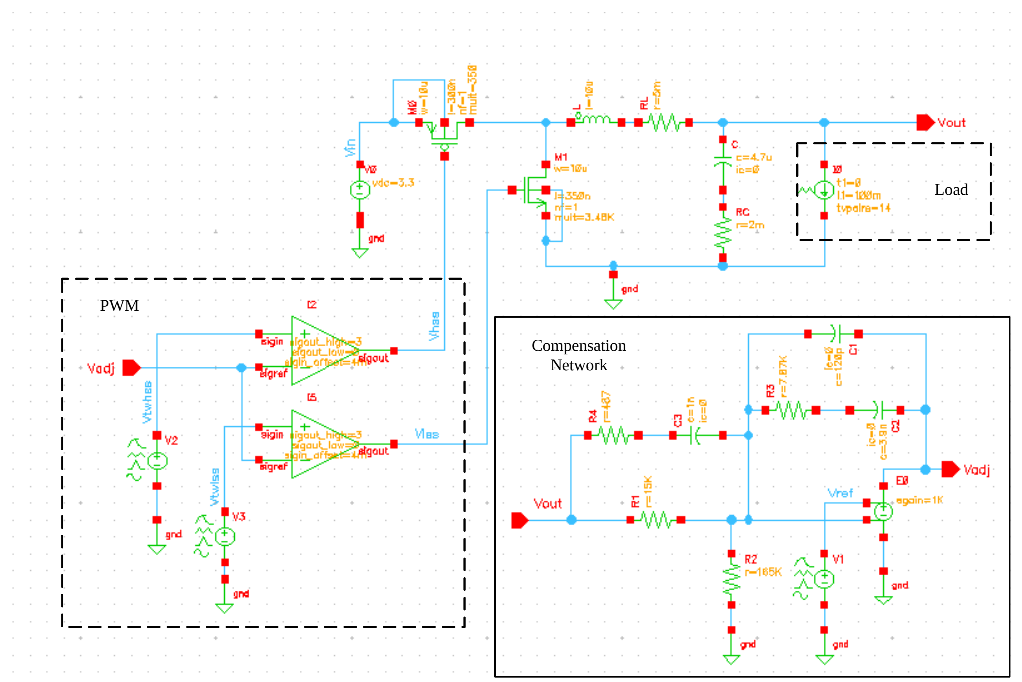

Figure 9: Synchronous DC-DC Buck Converter Schematic

Since the main focus of this report is on the corner effects of the PMOS HSS and NMOS LSS transistors, an ideal comparator and amplifier were used for the PWM and compensation network respectively.

3.3 Simulated Magnitude and Phase Plots

To verify that the compensation network will in fact produce the required phase margin by counteract the double pole of the LC filter, the magnitude and phase plots of each block where simulated. The 180 degree phase change around the LC double pole can clearly be seen in Figure 10. Likewise, the 180 degree phase boost from the Type-III compensation network is shown in Figure 11. To achieve the closed loop plot given in Figure 12, the LC filter and compensation network were wired together and the gain of the PWM stage (V_{IN} / V_{TW} = 3.3 V / 1.5 V) was multiplied into calculation. Also, the equivalent resistance (601 mΩ) of the HSS and LSS was added to R_L in the LC filter. The phase margin for this closed loop system is around 55.5 degrees. Also note that F_0 is at 62.77 kHz which is close to the desired value of 58.33 kHz. The difference is most likely due to adding the ideal gain of the PWM stage into the gain calculation.

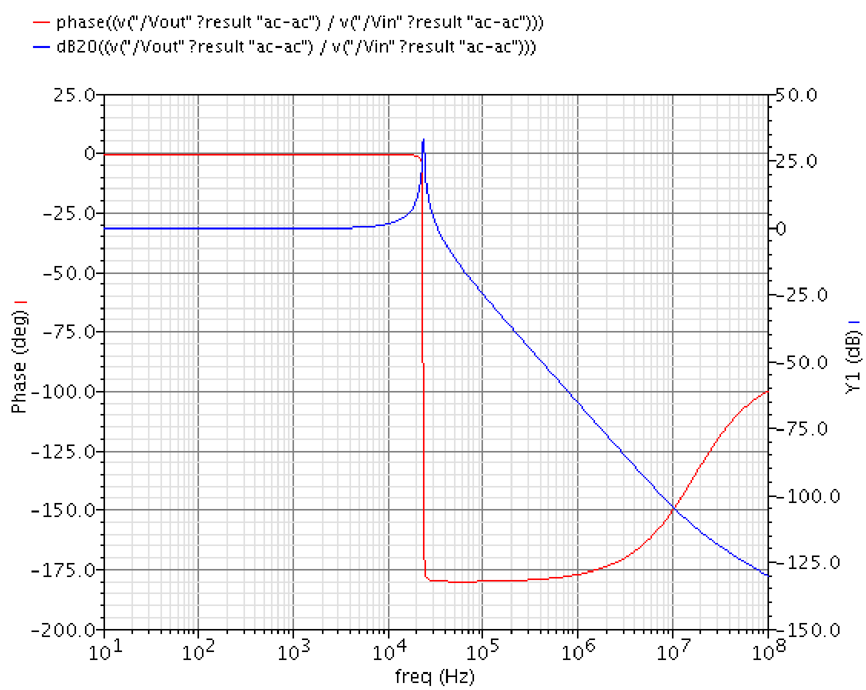

Figure 10: Simulated LC Filter Bode Plot

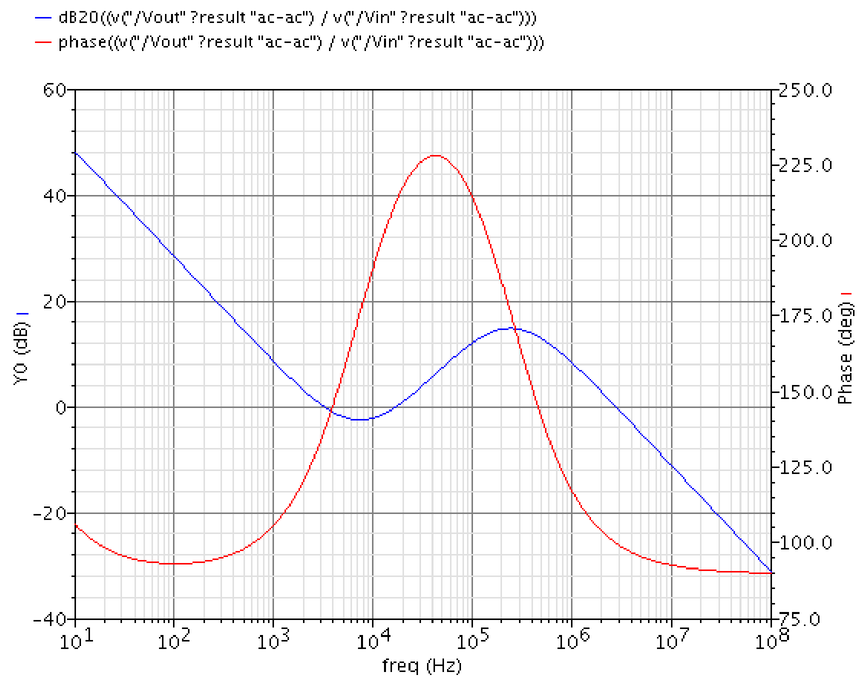

Figure 11: Simulated Type-III Compensation Network Bode Plot

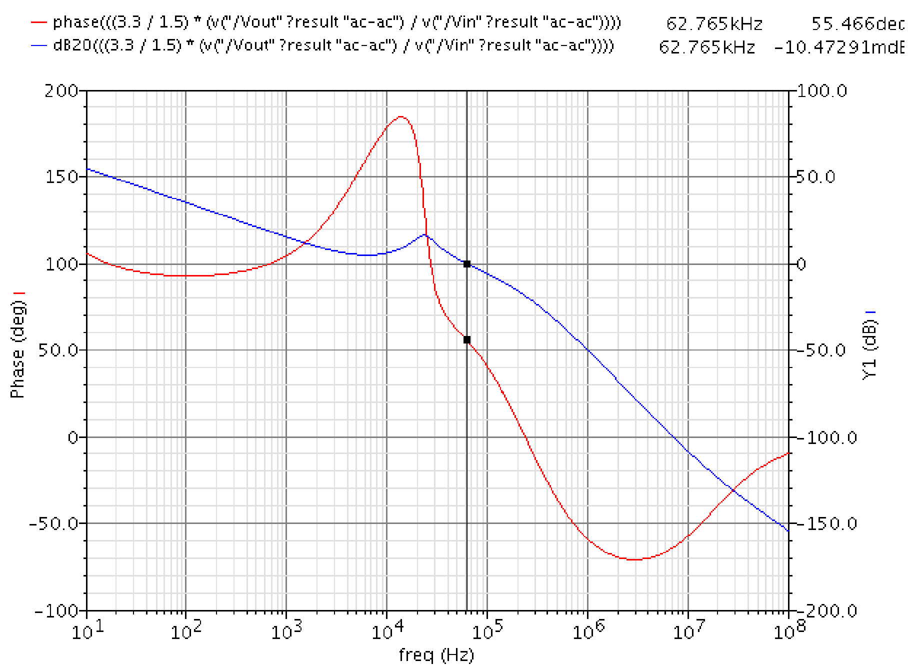

Figure 12: Simulated Closed Loop Bode Plot

3.4 Simulation Setup

The 0.13µ CMOS process corners, slow (ss), typical (tt), and fast (ff) are determined by the semiconductor fab based off of variations in their fabrication process. The automotive temperature range is -40 to 125 °C, with +27 °C being nominal. Most simulations are done over all corners for 100 mA load steps from 100 mA to 500 mA unless otherwise stated. For the efficiency measurements, average voltage and current levels (for 250 µs) were taken during a steady state output. The peak to peak measurements for the output voltage and inductor current were also sampled over a 250 µs window.

3.5 Start-up Response

The start-up response of the the synchronous buck converter at typical process, +27 °C, and nominal load current (I_{LOAD} = 200 mA) is shown in Figure 13. The inductor current (*/L/PLUS*) at start-up almost instantaneously rises just above 1.0 A, since the HSS transistor is on due to the output voltage being well below the reference voltage (V_{REF}). Also, V_{OUT} (*/Vout*) responds by jumping up to 2.35 V. Both of these values exceed the limits of this design, so a soft start-up mechanism is utilized. Soft start-up is accomplished by ramping V_{REF} from 0 to 1.65 V over 100 µs (see Figure 14). The start-up current through the inductor is reduced to 0.48 A, while V_{OUT} slowly ramps to the desired voltage.

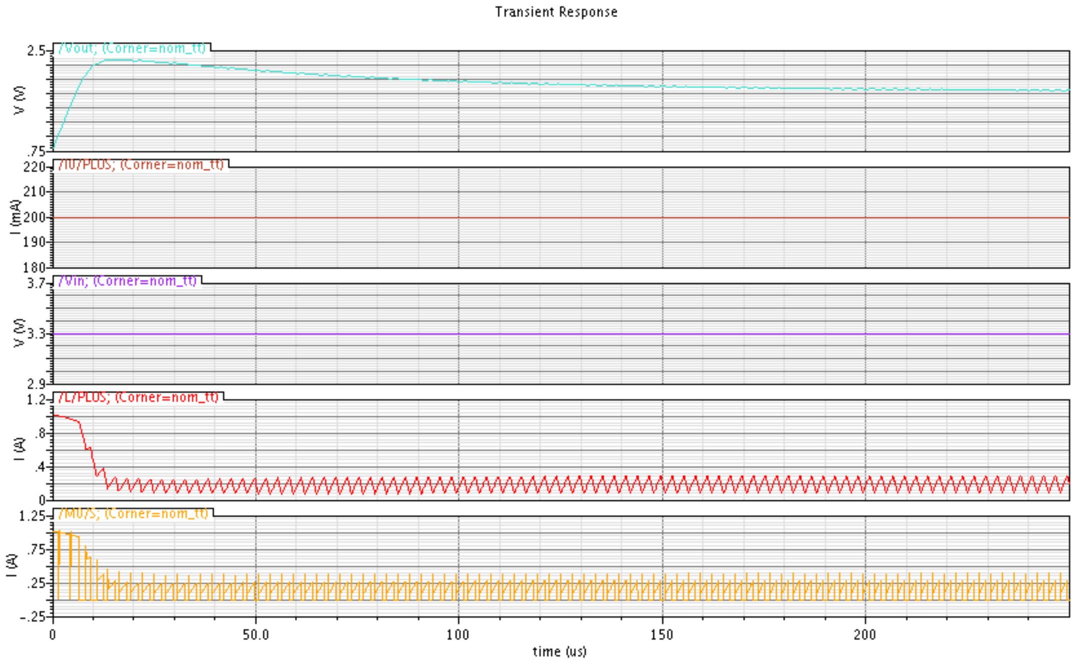

Figure 13: Start up Response (I_{LOAD} = 200 mA)

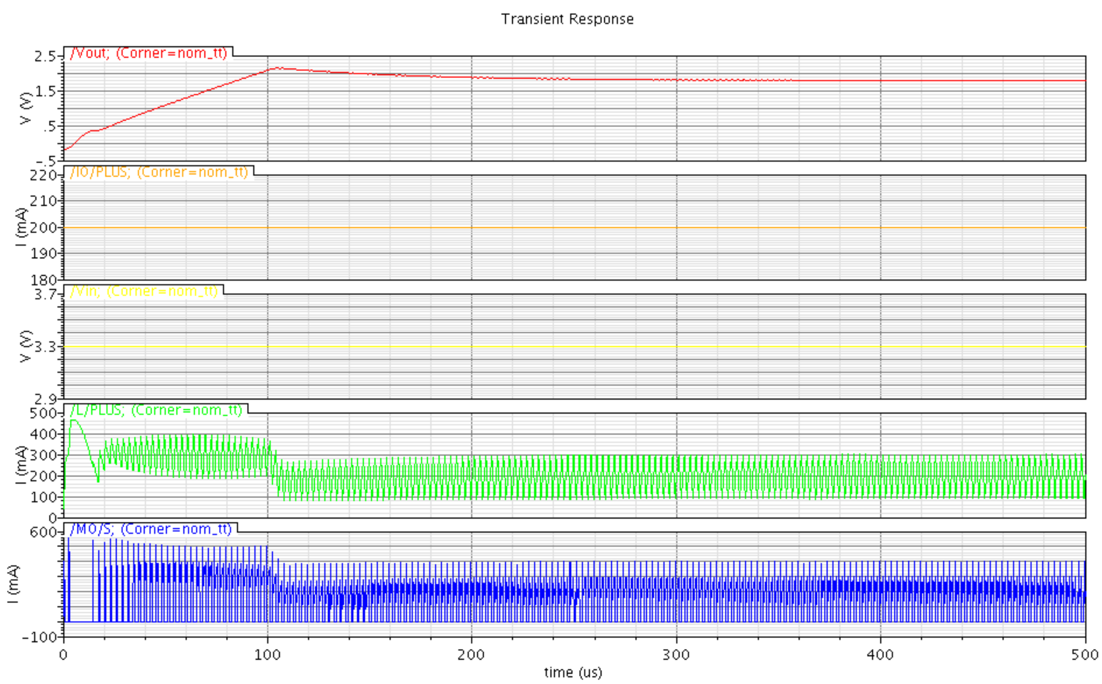

Figure 14: Start-up Response with 100 µs Soft Start (I_{LOAD} = 200 mA)

3.6 EFFICIENCY

The efficiency of the synchronous buck converter design over process, temperature, and load current is shown in Table 4 and Figure 15. The efficiency at typical process (tt) and nominal temperature (+27 °C) ranges from 80.86% to 95.56% at different loads. The maximum efficiency (96.55%) is at ff process, -40 °C, and I_{Load} = 100 mA and the minimum efficiency (73.50%) is at ss process, +125 °C, and I_{Load} = 500 mA.

Process Variation

The average percent change of efficiency from tt to ss for the same temperature and load is -1.14%, meaning that if at +27 °C and 100 mA the efficiency is 90%, the average efficiency at ss for the same temperature and load would be 0.90 + (-0.0114 * 0.90) = 88.97%. Likewise, the average percent change of efficiency from tt to ff for the same temperature and load is +0.91%.

Temperature Variation

The average percent change of efficiency from +27 °C to -40 °C for the same process and load is +2.17%. The average percent change of efficiency from +27 °C to +125 °C for the same process and load is -3.60%.

	Process	**ss**			**tt**			**ff**		
Load (mA)	**Temperature (°C)**	**-40**	**+27**	**+125**	**-40**	**+27**	**+125**	**-40**	**+27**	**+125**
100	**Efficiency (%)**	96.12	95.29	94.06	96.34	95.56	94.39	96.55	95.81	94.67
200		93.27	91.95	89.96	93.67	92.44	90.57	94.01	92.87	91.11
300		90.15	88.21	85.24	90.81	89.00	86.23	91.35	89.67	87.10
400		86.59	83.94	79.80	87.63	85.14	81.32	88.43	86.13	82.62
500		82.56	79.06	73.50	84.12	80.86	75.79	85.27	82.28	77.67

Table 4: Efficiency Data

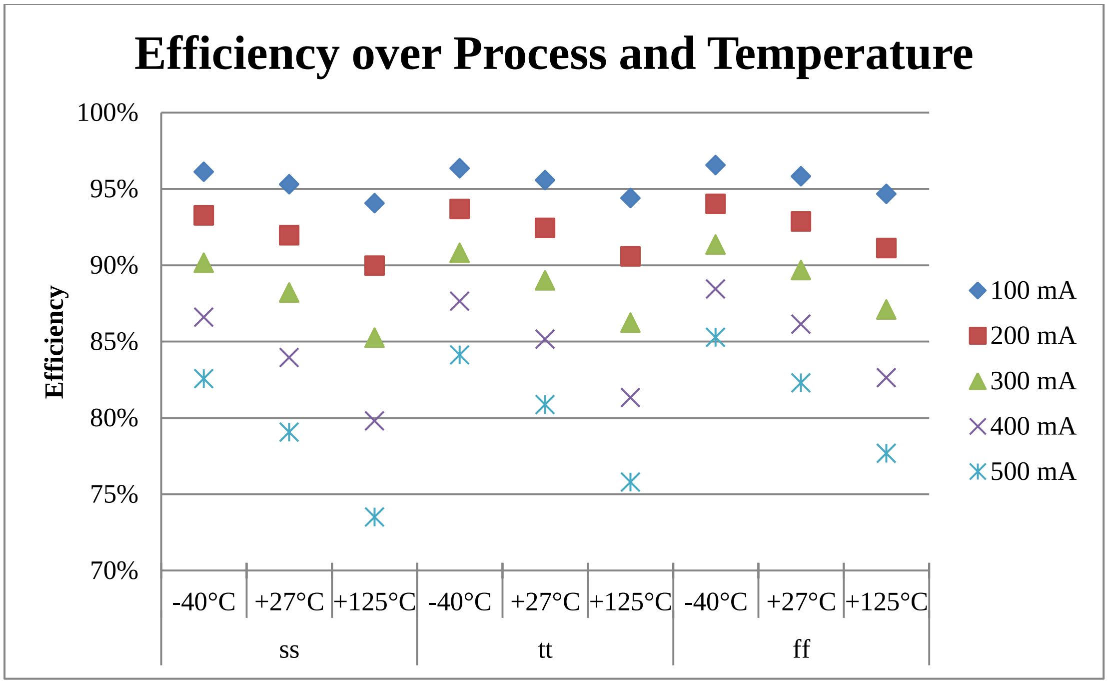

Figure 15: Efficiency Comparison

3.7 OUTPUT VOLTAGE RIPPLE

The output voltage (V_{OUT}) ripple over process, temperature, and load current is shown in Table 5 and Figure 16. The output ripple at typical process (tt) and nominal temperature (+27 °C) ranges 14.24 mV to 24.67 mV at different loads. The maximum output ripple (25.05 mV) occurs with ff process, -40 °C, and I_{Load} = 100 mA. The minimum output ripple (12.68 mV) occurs at ss process, +125 °C, and I_{Load} = 500 mA.

Process Variation

The average change in output voltage ripple from tt to ss for the same temperature and load is -0.89%. The average change in output voltage ripple from tt to ff for the same temperature and load is +0.81%.

Temperature Variation

The average change in output voltage ripple from +27 °C to -40 °C for the same process and load is +2.08%. The average change in output voltage ripple from +27 °C to +125 °C for the same process and load is -3.61%.

The waveform of the output ripple at I_{Load} = 200 mA is also shown in Figure 17.

	Process	**ss**			**tt**			**ff**		
Load (mA)	**Temperature (°C)**	**-40**	**+27**	**+125**	**-40**	**+27**	**+125**	**-40**	**+27**	**+125**
100	**V_{OUT} Ripple (mV)**	25.03	24.85	24.61	24.85	24.67	24.55	25.05	24.83	24.52
200		17.10	17.07	16.77	17.18	16.98	16.87	17.17	17.02	16.89
300		16.42	16.10	15.65	16.51	16.22	15.80	16.54	16.32	15.95
400		15.65	15.11	14.32	15.73	15.33	14.62	15.88	15.50	14.82
500		14.60	13.93	12.68	14.92	14.24	13.15	15.09	14.48	13.51

Table 5: V_{OUT} Ripple Data

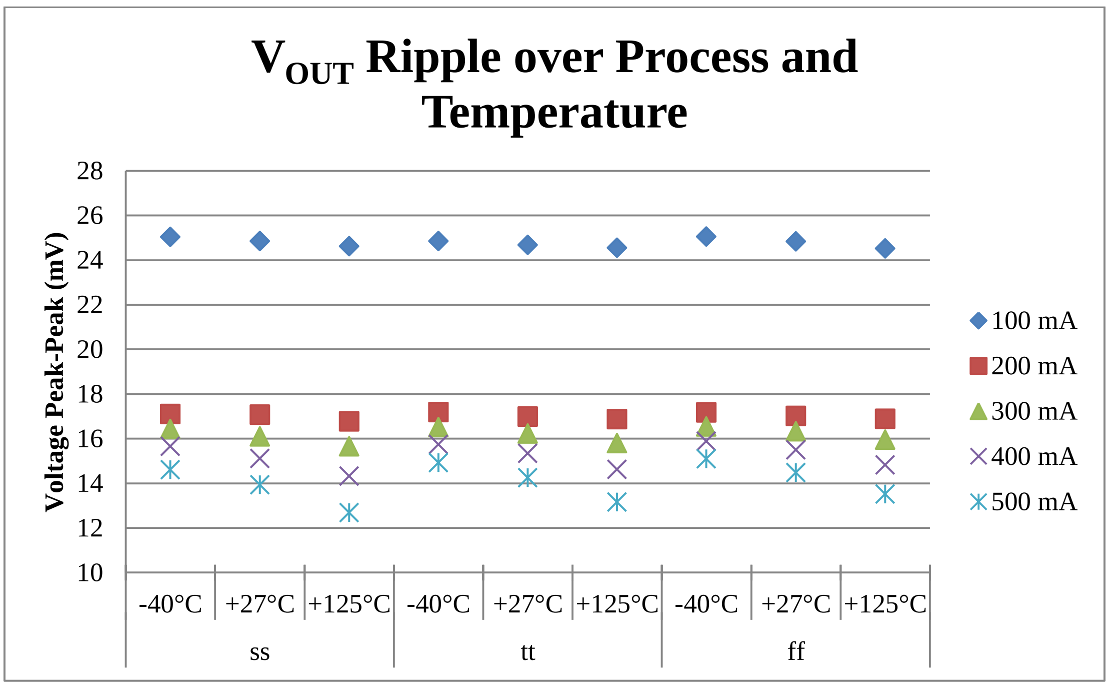

Figure 16: V_{OUT} Ripple Comparison

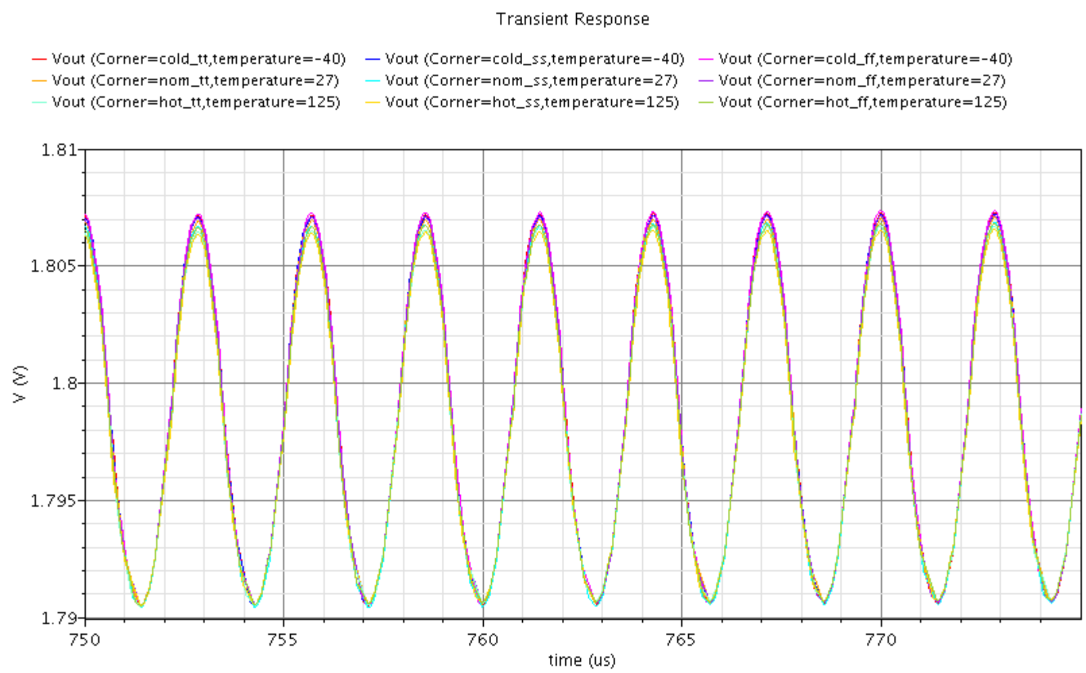

Figure 17: V_{OUT} Ripple Waveform (I_{Load} = 200 mA)

3.8 Inductor Peak Current

The inductor peak to peak current over process, temperature, and load current is shown in Table 6 and Figure 18. The inductor peak to peak current at typical process (tt) and nominal temperature (+27 °C) ranges from 171.7 mA to 225.7 mA at different loads. The maximum inductor peak to peak current (227.7 mA) occurs with ff process, -40 °C, and I_{Load} = 100 mA. This means that the converter is operating in discontinuous mode, where the current in the inductor goes to zero in one duty cycle due to the light load condition. The minimum inductor peak to peak current (137.3 mA) occurs at ss process, -40 °C, and I_{Load} = 500 mA.

Process Variation

The average change in inductor peak to peak current from tt to ss for the same temperature and load is -2.11%. The average change in inductor peak to peak current from tt to ff for the same temperature and load is +1.55%.

Temperature Variation

The average change in inductor peak to peak current from +27 °C to -40 °C for the same process and load is +3.35%. The average change in inductor peak to peak current from +27 °C to +125 °C for the same process and load is -6.60%.

A waveform of the output ripple at I_{Load} = 200 mA is also shown in Figure 19.

	Process	**ss**			**tt**			**ff**		
Load (mA)	**Temperature (°C)**	**-40**	**+27**	**+125**	**-40**	**+27**	**+125**	**-40**	**+27**	**+125**
100	**Inductor Peak to Peak Current (mA)**	226.9	225.1	222.5	227.2	225.7	223.2	227.7	226.2	224.1
200		217.7	214.1	208.0	218.8	215.4	209.9	219.7	216.5	211.5
300		207.2	200.7	190.4	209.2	203.3	194.0	211.0	205.5	197.0
400		194.4	184.6	167.7	198.0	189.1	174.2	200.8	192.7	179.5
500		178.7	164.0	137.3	184.8	171.7	148.9	189.0	177.5	157.7

Table 6: Inductor Peak to Peak Current Data

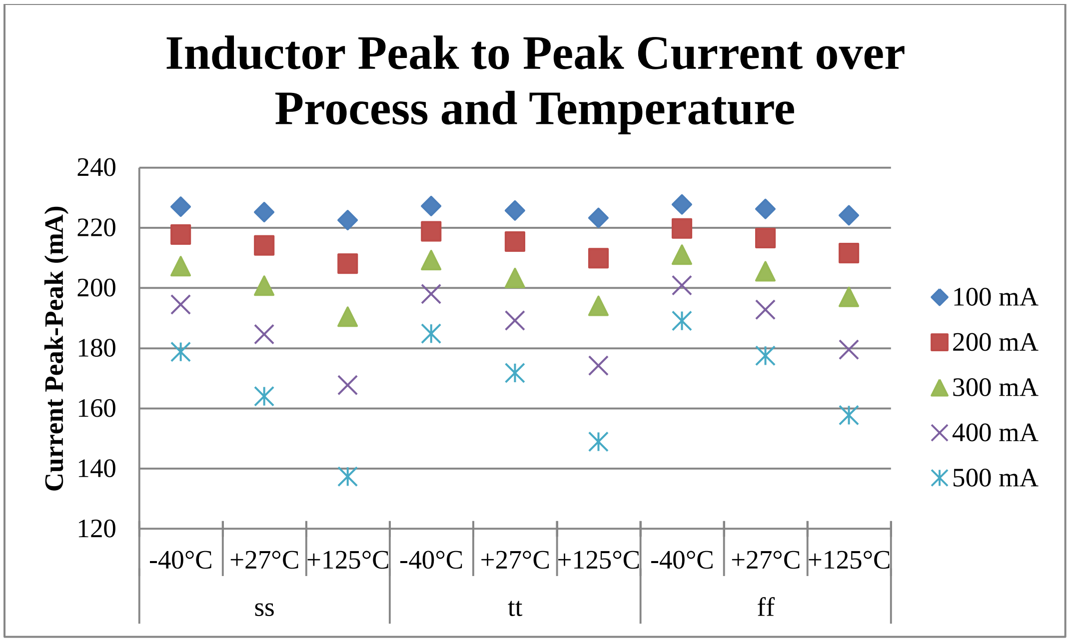

Figure 18: Inductor Peak to Peak Current Comparison

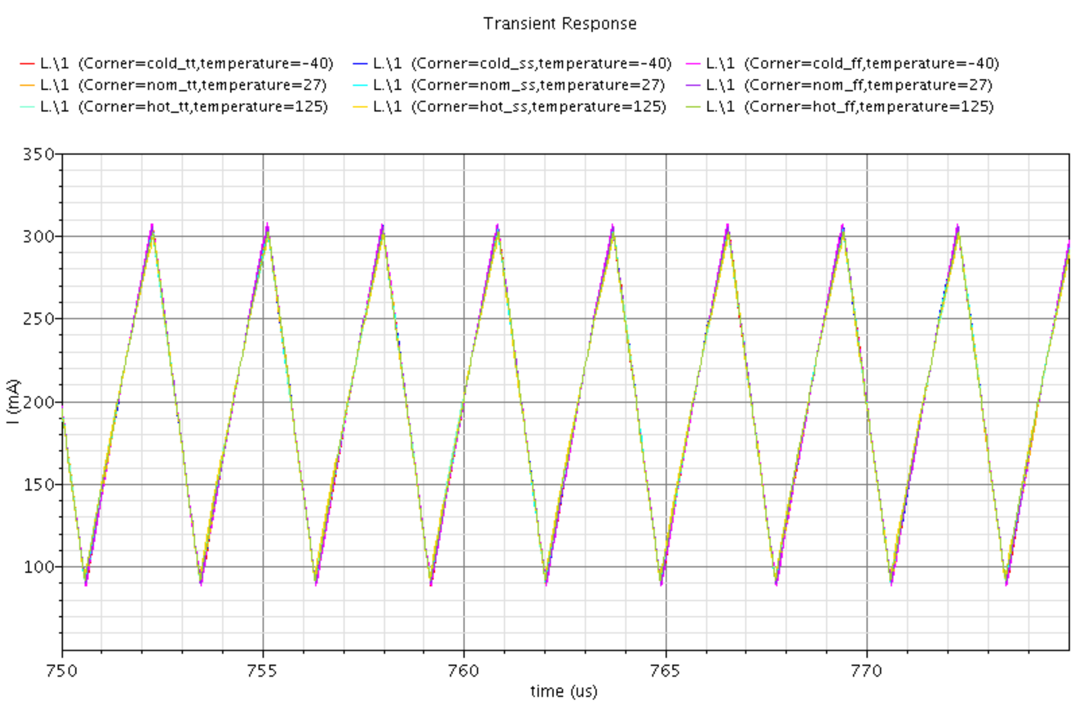

Figure 19: Inductor Peak to Peak Current Waveform (I_{Load} = 200 mA)

3.9 OUTPUT VOLTAGE LOAD RESPONSE (MIN TO MAX)

The output voltage (V_{OUT}) load response to an instantaneous load change from 100 mA to 500 mA over process and temperature is shown in Table 7 and Figure 20.

Process Variation

The average change in output voltage load response from tt to ss for the same temperature is +0.99%. The average output voltage load response change from tt to ff for the same temperature is -0.72%.

Temperature Variation

The average output voltage load response change from +27 °C to -40 °C for the same process is -1.91%. The average output voltage load response change from +27 °C to +125 °C for the same process is +2.93%.

A waveform of the output voltage response is also shown in Figure 21. The average response time it takes V_{OUT} to get back within 5% of the desired output voltage is 30 μs.

Process	**ss**			**tt**			**ff**		
Temperature (°C)	**-40**	**+27**	**+125**	**-40**	**+27**	**+125**	**-40**	**+27**	**+125**
Delta VOUT (mV)	-229	-233	-241	-227	-231	-238	-225	-230	-236

Table 7: V_{OUT} Min to Max Load Response

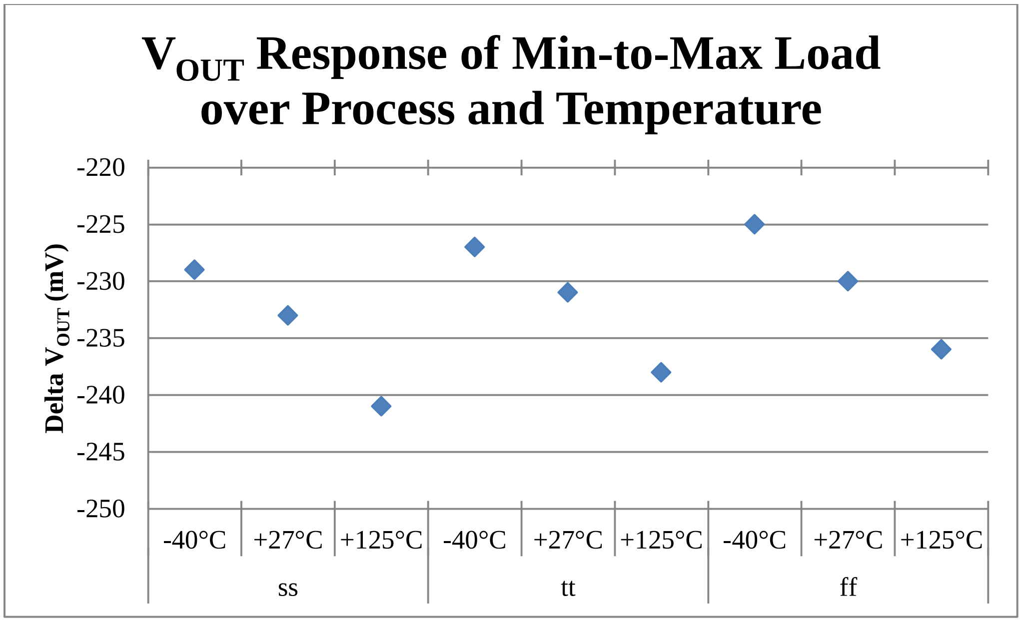

Figure 20: V_{OUT} Min to Max Load Response Comparison

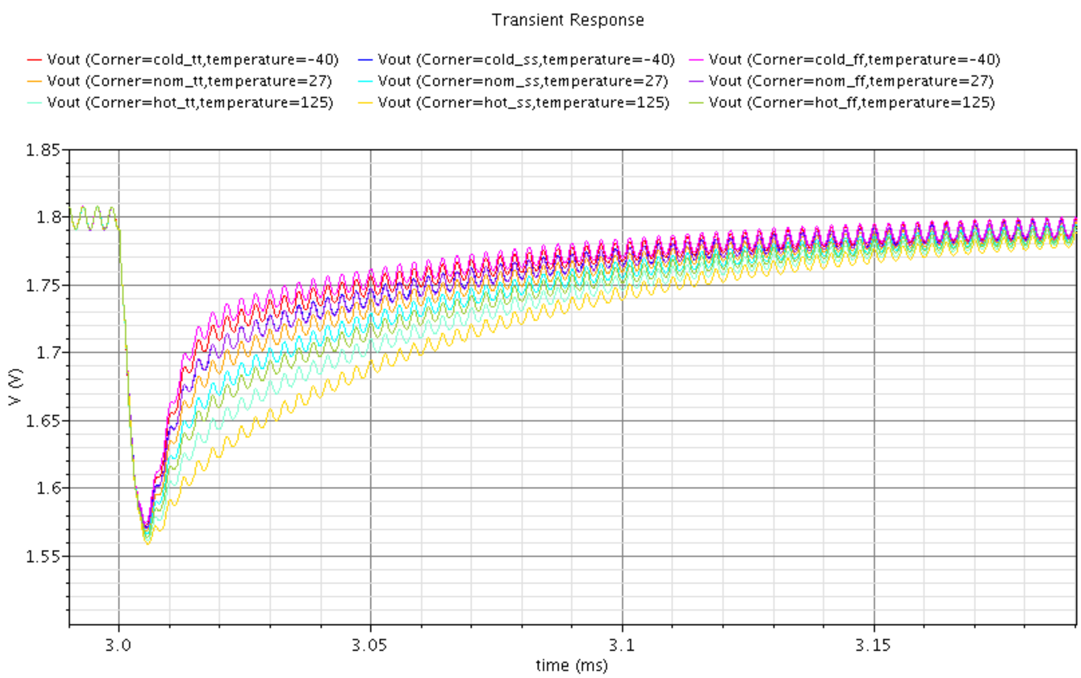

Figure 21: V_{OUT} Min to Max Load Response Waveform

3.10 OUTPUT VOLTAGE LOAD RESPONSE (MAX TO MIN)

The output voltage (V_{OUT}) load response to an instantaneous load change from 500 mA to 100 mA over process and temperature is shown in Table 8 and Figure 22.

Process Variation

The average output voltage load response change from tt to ss for the same temperature is -0.50%. The average output voltage load response change from tt to ff for the same temperature is -2.47%.

Temperature Variation

The average output voltage load response change from +27 °C to -40 °C for the same process is -1.99%. The average output voltage load response change from +27 °C to +125 °C for the same process is +4.47%.

A waveform of the output voltage response is also shown in Figure 23. The average response time it takes V_{OUT} to get back within 5% of the desired output voltage is 30 μs.

Process	**ss**			**tt**			**ff**		
Temperature (°C)	**-40**	**+27**	**+125**	**-40**	**+27**	**+125**	**-40**	**+27**	**+125**
Delta VOUT (mV)	237	242	249	234	239	259	233	237	244

Table 8: V_{OUT} Max to Min Load Response

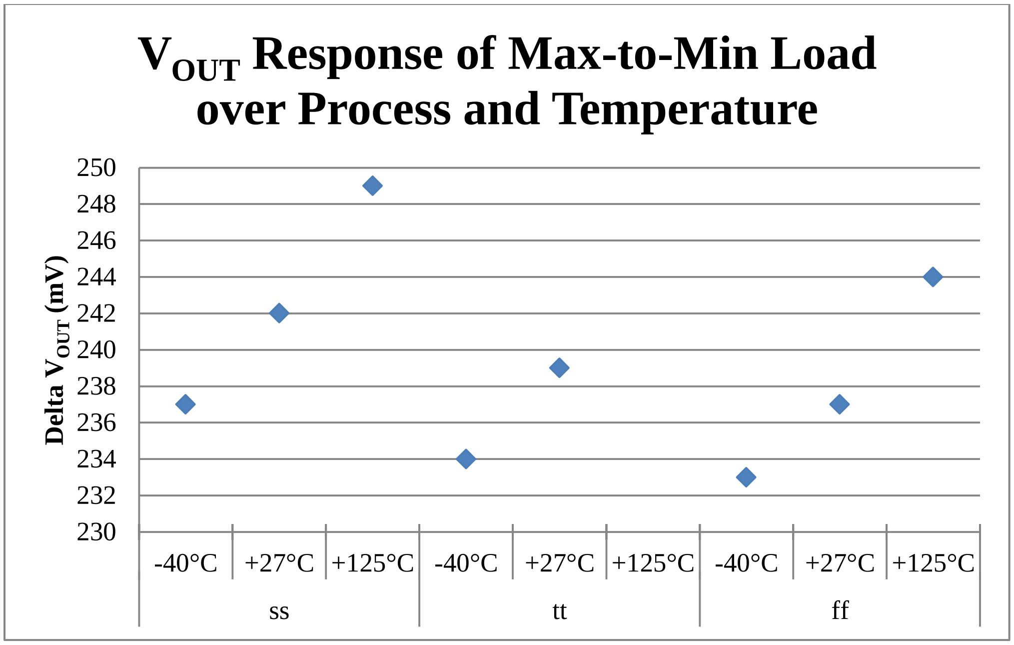

Figure 22: V_{OUT} Max to Min Load Response Comparison

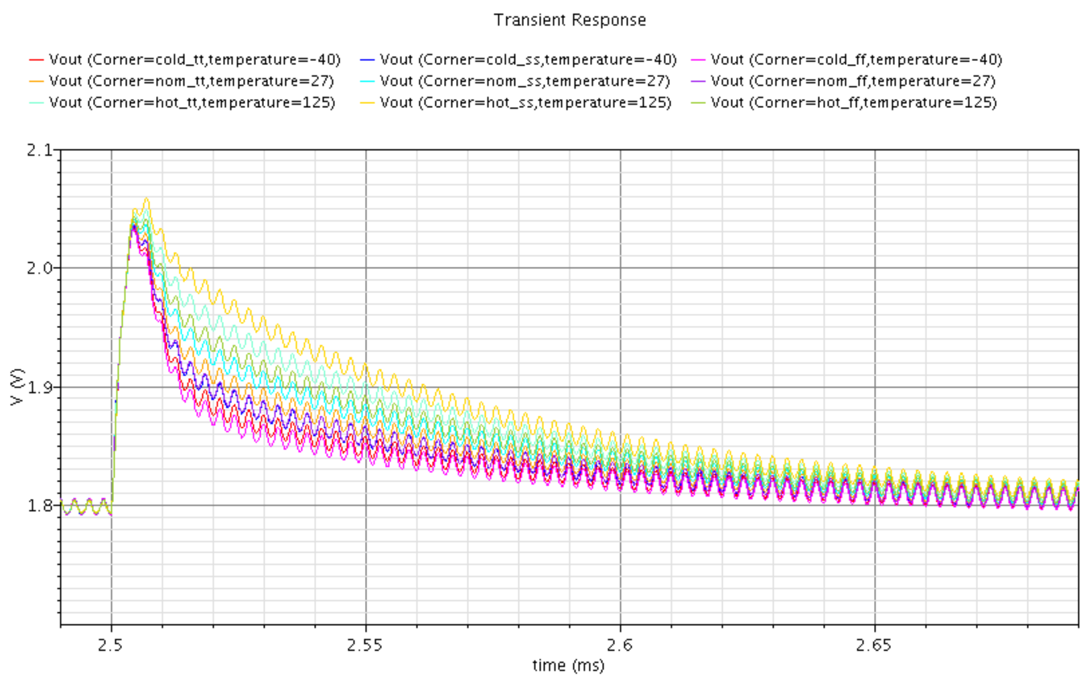

Figure 23: V_{OUT} Max to Min Load Response Waveform

Chapter 4: Conclusions

Designing a voltage-mode controlled synchronous buck converter is very challenging. The most difficult part is determining the compensation network and manipulating the poles and zeros to build a robust and balanced system. The PWM is a relatively simple concept, but a real world design of this block would be troublesome. Developing the triangle wave source and dead time scheme into the system can be accomplished by a good engineer, but having to take into account the shifts that occur because of process and temperature corners, ups the level of complexity.

The average measurement change for the 3.3 V to 1.8 V synchronous buck converter due to process and temperature corners is shown in Table 9 and Table 10 respectively. Changing temperature has a slightly greater impact on the value of each measurement than the process change does, most likely due to using the automotive temperature range over consumer (0 °C to +70 °C). The most volatile measurement over both process and temperature is inductor peak to peak current. This is mainly due to the direct correlation of the transistor switching properties (e.g. threshold voltage, $R_{DS,on}$, $I_{D,sat}$) to the amount of current available to the inductor during the charge and discharge states.

Measurement	**Slow Process (ss)**	**Fast Process (ff)**
Efficiency	-1.14%	+0.91%
V_{OUT} Ripple	-0.89%	+0.81%
Inductor Pk to Pk Current	-2.11%	+1.55%
V_{OUT} Load Response (Min to Max)	+0.99%	-0.72%
V_{OUT} Load Response (Max to Min)	-0.50%	-2.47%

Table 9: Average Measurement Change from Typical Process (tt)

Measurement	-40 °C	+125 °C
Efficiency	+2.17%	-3.60%
V_{OUT} Ripple	+2.08%	-3.61%
Inductor Pk to Pk Current	+3.35%	-6.60%
V_{OUT} Load Response (Min to Max)	-1.91%	+2.93%
V_{OUT} Load Response (Max to Min)	-1.99%	+4.47%

Table 10: Average Measurement Change from Nominal Temperature (+27 °C)

The V_{OUT} ripple measurement is related to the amount of peak to peak current through the inductor by $dV_{OUT} = (I * dt) / C$. The higher the output voltage ripple, the more unwanted electromagnetic emissions will be produced. At the lightest load (I_{Load} = 100 mA), the output ripple in this design is around 25 mV. This can be further reduced by increasing the amount of capacitance at the load, but then the feedback compensation network may lose some stability. Most power distribution networks on a printed circuit board will have some localized decoupling capacitors at the IC chip to help reduce spurious power supply voltage ripple.

The efficiency of the final design varies quite a bit over load, temperature, and process. Using the standard 0.13µ PMOS transistor as the current sourcing high-side switch has severe limitations. Increasing the size of the PMOS and NMOS transistors (by increasing the multiplier) allows for more current to be available for the output load (effectively lowering $R_{DS,on}$). Unfortunately, as size is increased, it will introduce a bigger capacitance, effectively needing to charge and discharge every cycle, leading to reduced efficiency. In order to obtain higher efficiency, a different technology will have to be explored (e.g. low $R_{DS,on}$ MOSFET).

Future work will focus on implementing the PWM and error amplifier in the commercial 0.13µ CMOS process along with the transistor switches. The effects of process and temperature corners will drastically increase as the more timing critical parameters of the PWM are implemented in silicon. Additional research will need to be conducted to increase the efficiency and reduce the unwanted output voltage ripple as well. Utilizing a multiphase design by replacing the external inductor with shared inductors or isolation transformers may also be another option to obtain a more robust synchronous buck converter design.

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Vita

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